DATA STRUCTURES MINI PROJECT

Library Management System

A simple demonstration of Stack (LIFO) and Queue (FIFO) using Java.

Java server interface

Manage books and student requests

Use the stack to process returned books and the queue to serve students in the order they arrive.

Returned Books

Stack · Last-In, First-Out (LIFO)

Book title

e.g. Data Structures in Java

Push Book

Pop / Process Latest

Refresh

Stack contents 0 books

Top of stack is shown first.

No returned books yet.

Student Waiting List

Queue · First-In, First-Out (FIFO)

Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 0 students

The first student in line is shown first.

No students waiting.

Data structures used

Stack — LIFO

The last book pushed onto the stack is the first book popped from it.

Queue — FIFO

The first student enqueued is the first student served and removed from the queue.

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Push Book

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Refresh

Stack contents 1 book

Top of stack is shown first.

01 Java Programming

Student Waiting List

Queue · First-In, First-Out (FIFO)

Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 0 students

The first student in line is shown first.

No students waiting.

Book added to the stack.

Data structures used

Stack — LIFO

The last book pushed onto the stack is the first book popped from it.

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Push Book

Pop / Process Latest

Refresh

Stack contents 2 books

Top of stack is shown first.

01 C Programming

02 Java Programming

Student Waiting List

Queue · First-In, First-Out (FIFO)

Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 0 students

The first student in line is shown first.

No students waiting.

Book added to the stack.

Data structures used

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Push Book

Pop / Process Latest

Refresh

Stack contents 3 books

Top of stack is shown first.

01 python

02 C Programming

03 Java Programming

Student Waiting List

Queue · First-In, First-Out (FIFO)

Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 0 students

The first student in line is shown first.

No students waiting.

Book added to the stack.

Data structures used

Stack — LIFO

The last book pushed onto the stack is the first book popped from it.

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Book title

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Push Book

Pop / Process Latest

Refresh

Stack contents 2 books

Top of stack is shown first.

01 C Programming

02 Java Programming

Student Waiting List

Queue · First-In, First-Out (FIFO)

Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 0 students

The first student in line is shown first.

No students waiting.

Processed: python

Data structures used

Stack — LIFO

The last book pushed onto the stack is the first book popped from it.

Queue — FIFO

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Push Book

Pop / Process Latest

Refresh

Stack contents 2 books

Top of stack is shown first.

01 C Programming

02 Java Programming

Student Waiting List

Queue · First-In, First-Out (FIFO)

Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 1 student

The first student in line is shown first.

01 Anu

Student added to the waiting queue.

Data structures used

Stack — LIFO

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Stack contents 2 books

Top of stack is shown first.

01 C Programming

02 Java Programming

Student Waiting List

Queue · First-In, First-Out (FIFO)

Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 2 students

The first student in line is shown first.

01 Anu

02 binu

Student added to the waiting queue.

Data structures used

Stack — LIFO

The last book pushed onto the stack is the first book popped from it.

Queue — FIFO

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Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 3 students

The first student in line is shown first.

01 Anu

02 binu

03 minu

Student added to the waiting queue.

Data structures used

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Student name

e.g. Anu Thomas

Enqueue

Dequeue / Serve Next

Refresh

Queue contents 2 students

The first student in line is shown first.

01 binu

02 minu

Now serving: Anu

Data structures used

Stack — LIFO

The last book pushed onto the stack is the first book popped from it.

Queue — FIFO

The first student enqueued is the first student served and removed from the queue.